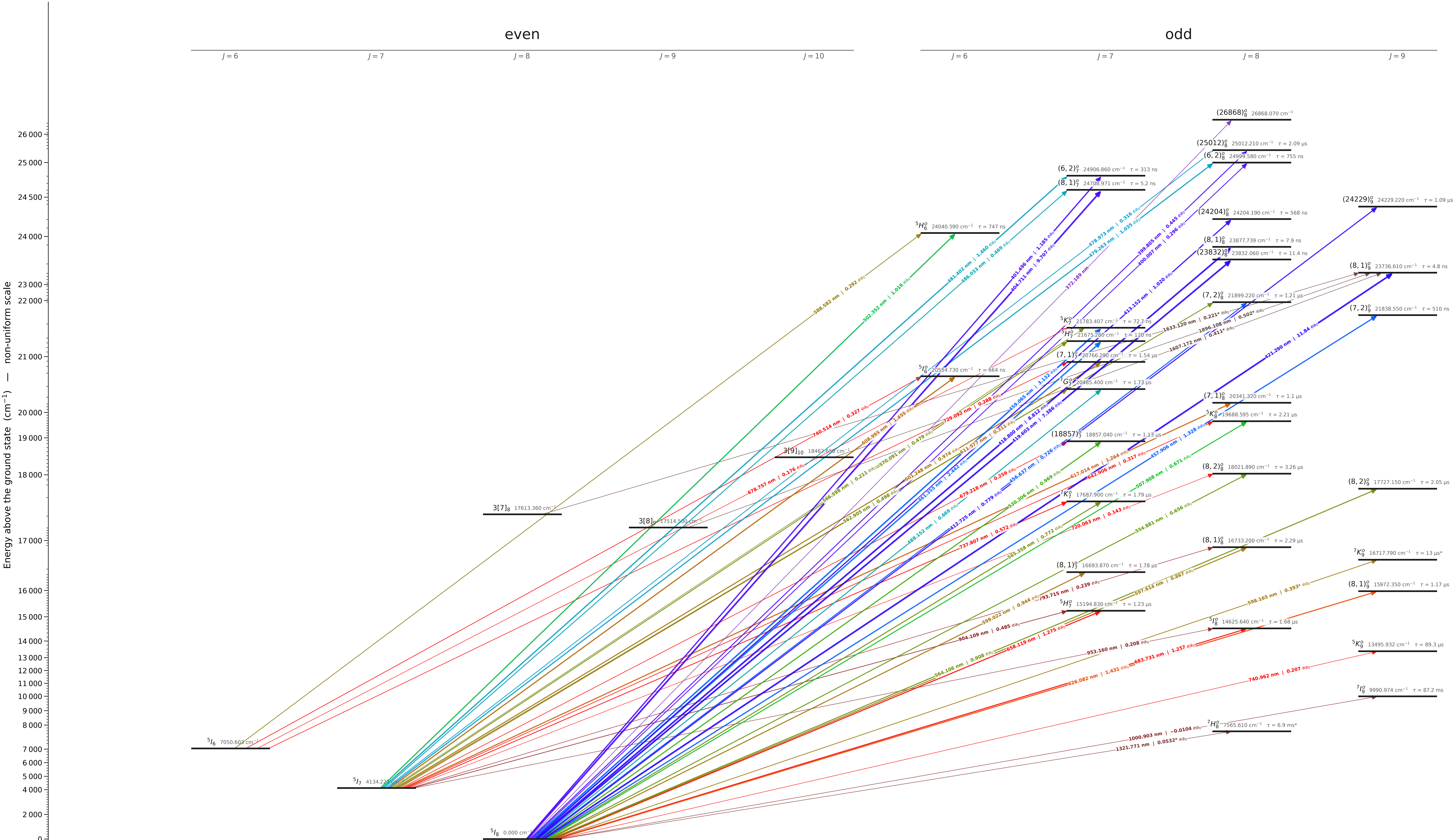
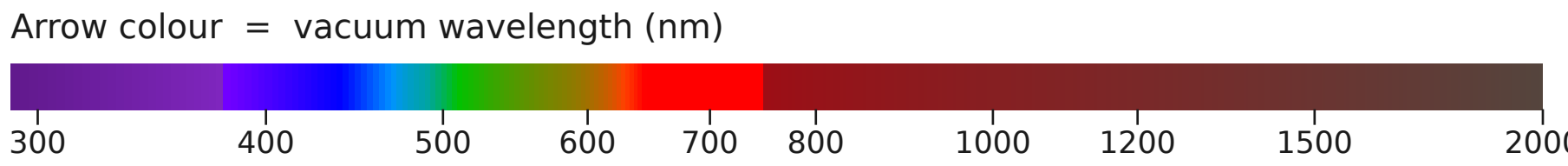


# Dysprosium-164

<sup>164</sup>Dy (Dy II), nuclear spin *I* = 0. Energy levels and transitions below 2 μm: 39 levels, 55 lines.

Dy<sup>+</sup> ionisation limit 47901.76 cm<sup>-1</sup> = 5.93906 eV (208.761 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|U||e r||J\rangle|$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.  
Level caption: NIST energy and measured lifetime (where one exists).



## Key transitions

| Transition                                                 | λ vac (nm) | λ air (nm) | ν (THz)       | [U  e r  J'] (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|------------------------------------------------------------|------------|------------|---------------|---------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| <sup>5</sup> I <sub>8</sub> - (26868)° <sub>8</sub>        | 372.1890   | 372.0832   | 805.4845      | -                               | -                    | -              | -          | UV ground-state line                                         |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>7</sub>         | 404.7113   | 404.5970   | 740.7563      | 9.707                           | 1.920e+08            | 30.61          | 99.8       | strong blue J=8->7 line                                      |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>         | 418.8001   | 418.6821   | 715.8366      | 8.812                           | 1.260e+08            | 20.15          | 99.5       | strong blue J=8->8 line                                      |
| <sup>5</sup> I <sub>8</sub> - (23832)° <sub>8</sub>        | 419.6028   | 419.4846   | 714.4672      | 7.386                           | 8.800e+07            | 13.96          | 100        | strong blue J=8->8 line                                      |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>         | 421.2902   | 421.1715   | 711.6057      | 11.84                           | 2.000e+08            | 33.16          | 100        | main broad cooling line: Zeeman slower, transverse coolin... |
| <sup>5</sup> I <sub>8</sub> - (7, 2)° <sub>8</sub>         | 456.6373   | 456.5093   | 656.5221      | 0.726                           | 6.600e+05            | 0.1315         | 79.9       | pure 6s->6p reference line for King plots                    |
| <sup>5</sup> I <sub>8</sub> - (8, 2)° <sub>8</sub>         | 564.1065   | 563.9499   | 531.4466      | 0.908                           | 4.900e+05            | 0.07764        | 100        | closed J=8->9 narrow line                                    |
| <sup>5</sup> I <sub>8</sub> - <sup>7</sup> K° <sub>8</sub> | 598.1652   | 597.9995   | 501.1867      | 0.393*                          | 7.700e+04            | 0.01224        | 100        | "598 nm" closed J=8->9 narrow line proposed for narrow-li... |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>         | 599.0223   | 598.8564   | 500.4696      | 0.944                           | 5.600e+05            | 0.08941        | 99.7       | narrow J=8->7 probe line                                     |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>         | 626.0819   | 625.9088   | 478.8390      | 1.431                           | 8.900e+05            | 0.136          | 100        | narrow-line MOT                                              |
| <sup>5</sup> I <sub>8</sub> - <sup>5</sup> H° <sub>8</sub> | 658.1186   | 657.9368   | 455.5295      | 1.275                           | 7.700e+05            | 0.1294         | 94.7       | narrow probe line used for Doppler thermometry of the 421... |
| <sup>5</sup> I <sub>8</sub> - <sup>5</sup> I° <sub>8</sub> | 683.7308   | 683.5421   | 438.4657      | 1.257                           | 5.893e+05            | 0.09474        | 99         | narrow J=8->8 optical pumping / demagnetization cooling l... |
| <sup>5</sup> I <sub>8</sub> - <sup>5</sup> K° <sub>8</sub> | 740.9619   | 740.7579   | 404.59900(10) | 0.207                           | 1.120e+04            | 0.001782       | 100        | ultra-narrow-line MOT / narrow-line cooling in tweezers      |
| <sup>5</sup> I <sub>8</sub> - <sup>7</sup> I° <sub>8</sub> | 1000.9034  | 1000.6291  | 299.5219      | 0.010                           | 1.150e+01            | 1.825e-06      | 100        | ultra-narrow                                                 |
| <sup>5</sup> I <sub>8</sub> - <sup>7</sup> H° <sub>8</sub> | 1321.7705  | 1321.4091  | 226.8113      | 0.053*                          | 1.460e+02            | 2.307e-05      | 100        | proposed ultra-narrow telecom-band line                      |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## Isotope shifts

| Transition                                          | λ vac (nm) | Isotopes | Shift (MHz) |
|-----------------------------------------------------|------------|----------|-------------|
| <sup>5</sup> I <sub>8</sub> - (26868)° <sub>8</sub> | 372.189    | 164-163  | -44(10)     |
| <sup>5</sup> I <sub>8</sub> - (26868)° <sub>8</sub> | 372.189    | 164-162  | -178(13)    |
| <sup>5</sup> I <sub>8</sub> - (26868)° <sub>8</sub> | 372.189    | 164-161  | -194(13)    |
| <sup>5</sup> I <sub>8</sub> - (26868)° <sub>8</sub> | 372.189    | 164-160  | -339(13)    |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>  | 421.290    | 164-163  | -616.3(5)   |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>  | 421.290    | 164-162  | -913.2(8)   |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>  | 421.290    | 164-161  | -1635.0(10) |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>  | 421.290    | 164-160  | -1895.0(20) |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>  | 421.290    | 164-158  | -2868(9)    |
| <sup>5</sup> I <sub>8</sub> - (8, 1)° <sub>8</sub>  | 421.290    | 164-156  | -4300(15)   |
| <sup>5</sup> I <sub>8</sub> - (7, 2)° <sub>8</sub>  | 456.637    | 164-163  | -660(3)     |
| <sup>5</sup> I <sub>8</sub> - (7, 2)° <sub>8</sub>  | 456.637    | 164-162  | -971.0(20)  |
| <sup>5</sup> I <sub>8</sub> - (7, 2)° <sub>8</sub>  | 456.637    | 164-161  | -1744(3)    |
| <sup>5</sup> I <sub>8</sub> - (7, 2)° <sub>8</sub>  | 456.637    | 164-160  | -2020(3)    |
| <sup>5</sup> I <sub>8</sub> - (7, 2)° <sub>8</sub>  | 456.637    | 164-158  | -3061(4)    |
| <sup>5</sup> I <sub>8</sub> - (7, 2)° <sub>8</sub>  | 456.637    | 164-156  | -4604(5)    |
| <sup>5</sup> I <sub>8</sub> - (8, 2)° <sub>8</sub>  | 564.106    | 164-163  | -685.0(20)  |
| <sup>5</sup> I <sub>8</sub> - (8, 2)° <sub>8</sub>  | 564.106    | 163-162  | -328(3)     |
| <sup>5</sup> I <sub>8</sub> - (8, 2)° <sub>8</sub>  | 564.106    | 162-161  | -801(3)     |
| <sup>5</sup> I <sub>8</sub> - (8, 2)° <sub>8</sub>  | 564.106    | 161-160  | -284.0(20)  |
| <sup>5</sup> I <sub>8</sub> - (8, 2)° <sub>8</sub>  | 564.106    | 164-162  | -1012.0(20) |
| <sup>5</sup> I <sub>8</sub> - (8, 2)° <sub>8</sub>  | 564.106    | 162-160  | -1086(5)    |

Shift = ν(first isotope) - ν(second isotope).